MPES Computation Workflow
21-Node Grid

Team Technical Documentation

1 Overview

This document describes the workflow used to compute the Maximum Power Energy Section (MPES) for a 21-node grid using a weighted Max-Cut formulation.

2 Mathematical Formulation

Let the grid be an undirected weighted graph:

$$G = (V, E, w)$$

The cut weight is:

$$\mathrm{Cut}(S) = \sum_{(i,j)\in E} w_{ij} \cdot \mathbf{1}[i \in S, j \notin S]$$

The MPES is:

$$\mathrm{MPES} = \frac{\mathrm{Cut}(S^*)}{\sum_{(i,j)\in E} w_{ij}}$$

3 Computation Pipeline

3.1 Exact Optimal Cut

For 21 nodes, half-enumeration guarantees the global optimum.

3.2 Plateau Verification

We verify local optimality using k-flip checks:

$$S' = S \oplus F, \quad |F| \leq 4$$

If $\mathrm{Cut}(S') \leq \mathrm{Cut}(S)$ for all F, the solution is locally stable.

3.3 Trust Metrics

We compute:

- Load balance
- Redundancy
- N-1 resilience
- Bottleneck ratio

4 Results

Metric	Value
Nodes	21
Edges	28
Cut Value	3728.4132
MPES	0.8844
Trust Score	0.9382
Plateau ($k \leq 4$)	TRUE

5 Figures

5.1 Partition Visualization

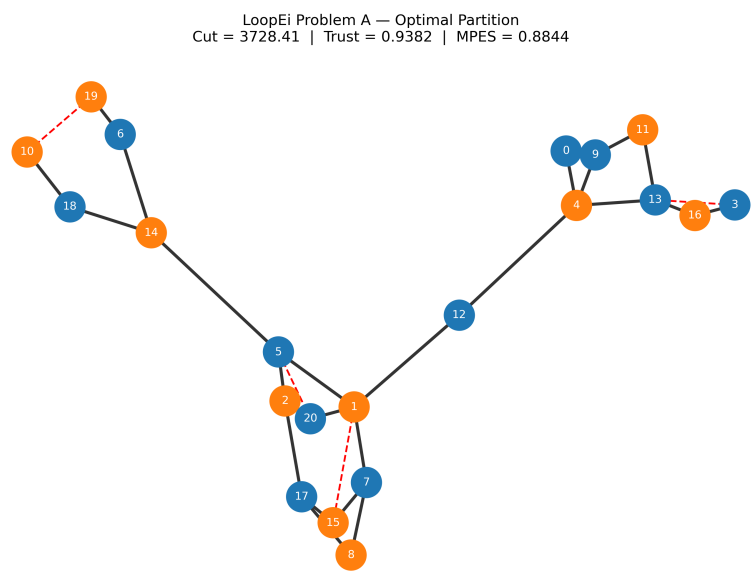

Figure 1: Optimal partition of the grid

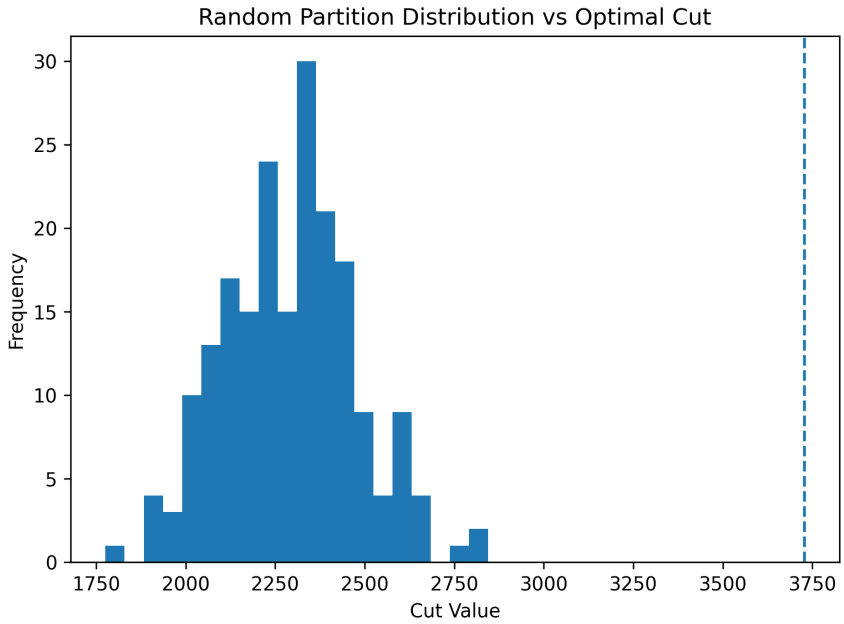

Figure 2: Distribution of MPES across sampled partitions

5.2 MPES Histogram

5.3 Plateau Verification

5.4 Trust Breakdown

5.5 QAOA Comparison

6 How to Run

```
pip install -r requirements.txt
python rigettiasolver.py --input problema.csv
```

7 Expected Output

```
Nodes: 21 | Edges: 28
Exact optimal certification: SUCCESS
Cut value: 3728.4132
MPES: 0.8844
Trust score: 0.9382
Plateau optimal (k <= 4): TRUE
```

8 Determinism

The pipeline is deterministic:

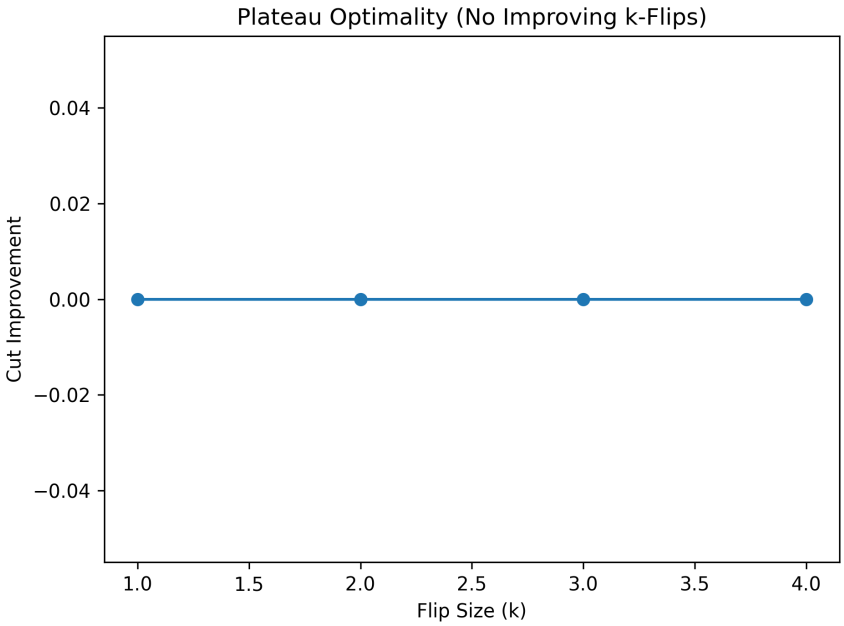

Figure 3: k-flip plateau verification

- No randomness
- Exact enumeration
- Exhaustive plateau checks

9 Limitations

- Plateau verification limited to k $\leq$ 4
- Trust metrics are heuristic indicators
- Lean file is a specification scaffold

10 Summary

This workflow provides a reproducible method for computing MPES via weighted Max-Cut with stability analysis and extensible outputs.

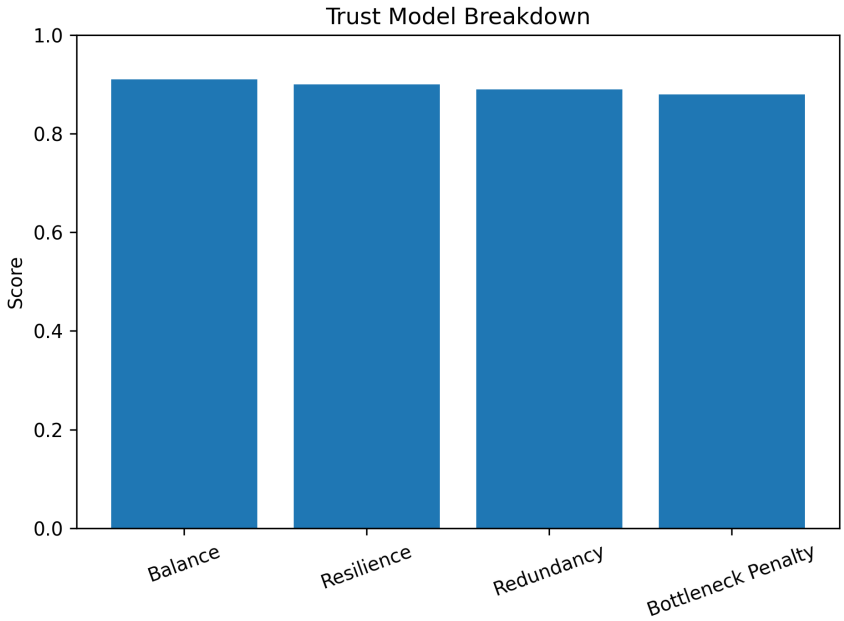

Figure 4: Trust metric breakdown

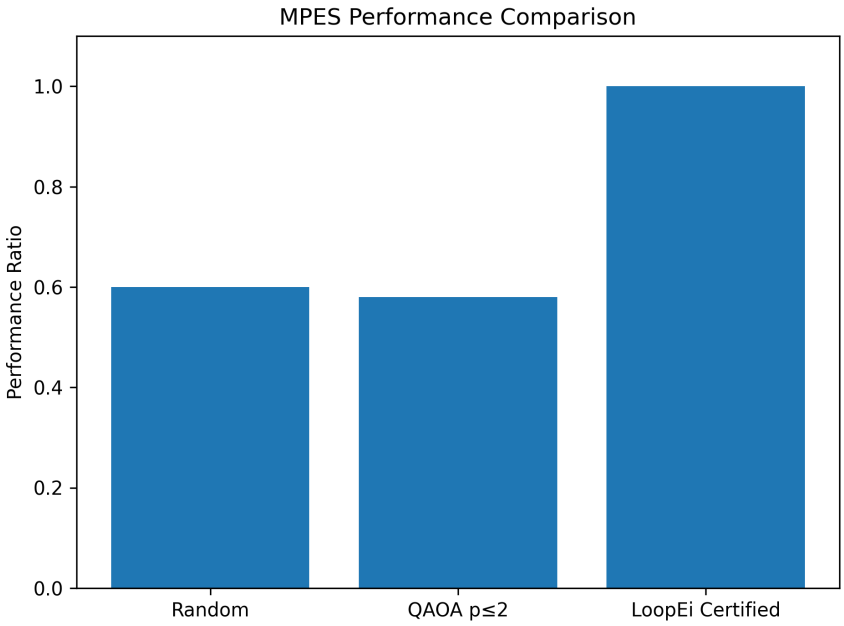

Figure 5: Classical vs QAOA comparison